

Week 34: Linear Regression

From data to the normal equations, and to the statistics of the estimator

With pauses for questions

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education,
University of Oslo, Norway

Plan

- ▶ Why linear regression at all
- ▶ Inputs, outputs, and the design matrix; a polynomial of degree five as running example
- ▶ The cost function and the normal equations
- ▶ Geometry: projection and the hat matrix
- ▶ Never invert $\mathbf{X}^T \mathbf{X}$: how to compute it
- ▶ Measuring quality: MSE and R^2
- ▶ Weighted least squares and the χ^2 function
- ▶ Least squares is maximum likelihood
- ▶ Mean, variance and confidence intervals of $\hat{\theta}$; what collinearity does

How the slides work

Every few slides there is a *Pause and think* frame. Talk to your neighbour for a minute; then we discuss. Reading: Chapter 3, Sections 3.1–3.7.

Why linear regression (aka ordinary least squares and family)?

Fitting a continuous function with a *linear parametrisation* in the parameters θ .

- ▶ Method of choice for fitting a continuous function
- ▶ Analytical expressions for the parameters *and* for their statistical properties: mean values, variances, confidence intervals
- ▶ Analytical relation with a probabilistic interpretation (maximum likelihood, Bayes)
- ▶ The natural place to meet bias–variance, cross-validation, resampling and regularisation – everything else in this course reuses them
- ▶ Easy to code, and links directly to logistic regression, neural networks, support vector machines and gradient descent

Further reading: van Wieringen, *Lecture notes on ridge regression* (arXiv:1509.09169); Mehta et al., *A high-bias, low-variance introduction to machine learning for physicists* (arXiv:1803.08823).

Regression analysis: the overarching aim

Inputs and outputs

Regression describes how the distribution of a random variable y – the *output* or response – varies with a set of variables $\mathbf{x} = [x_0, x_1, \dots, x_{p-1}]$ – the *inputs*, features or predictors. We shall just say *inputs* and *outputs*.

A regression model aims at the conditional distribution $p(\mathbf{y} \mid \mathbf{x})$, or more traditionally at a function $\mathbf{y}(\mathbf{x})$, estimated from a data set with

- ▶ n cases $i = 0, 1, \dots, n - 1$,
- ▶ one output y_i per case,
- ▶ p inputs per case, $\mathbf{x}_i = [x_{i0}, x_{i1}, \dots, x_{i,p-1}]$.

Collected in a matrix, the inputs form the *design matrix* $\mathbf{X} \in \mathbb{R}^{n \times p}$: one row per case, one column per feature. The goal is to exploit the relationship between $\mathbf{y}$ and $\mathbf{X}$ to make predictions, infer dependencies, and quantify how sure we are.

Example: a polynomial of degree five

Our running example: $n = 100$ points x_i on $[0, 1]$, targets from a smooth function plus noise, fitted with a polynomial of degree 5,

$$\tilde{y}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 + \theta_5 x^5.$$

Six predictors – the intercept and the powers $x, \dots, x^5$ – and one row per data point:

```
n = 100
x = np.linspace(0, 1, n)
y = 2.0 + 3.0*x - 5.0*x**2 + np.exp(-x) + 0.1*np.random.randn(n) # "the data"

X = np.zeros((n, 6))
for j in range(6):
    X[:, j] = x**j # columns 1, x, x^2, ..., x^5
# or in one line: X = np.vander(x, 6, increasing=True)
```

Here the predictors are functions we chose. In the credit-card default data (Taiwan, $n = 30\,000$ card holders, 24 attributes) they come from the data base – same mathematics, different origin of the columns.

Pause and think 1: shapes

Question

For the degree-5 example: what are n and p ? What is the shape of $\mathbf{X}$, of θ , of $\mathbf{X}^T \mathbf{X}$, of $\mathbf{X} \mathbf{X}^T$? And what changes if we go to degree 10 – and if we then have only $n = 8$ data points?

Pause and think 1: shapes

Question

For the degree-5 example: what are n and p ? What is the shape of $\mathbf{X}$, of $\boldsymbol{\theta}$, of $\mathbf{X}^T \mathbf{X}$, of $\mathbf{X} \mathbf{X}^T$? And what changes if we go to degree 10 – and if we then have only $n = 8$ data points?

Answer

$n = 100$, $p = 6$ (degree 5 plus the intercept). So $\mathbf{X} \in \mathbb{R}^{100 \times 6}$, $\boldsymbol{\theta} \in \mathbb{R}^6$, $\mathbf{X}^T \mathbf{X}$ is 6×6 – a small matrix however large n is – while $\mathbf{X} \mathbf{X}^T$ is 100×100 . Degree 10 adds five *columns*, $p = 11$; the model is still linear in $\boldsymbol{\theta}$. With $n = 8 < p = 11$ we have more unknowns than equations: infinitely many polynomials pass through all eight points, and $\mathbf{X}^T \mathbf{X}$ is singular – keep this in mind for the next slides.

A fitting problem written as linear algebra

Data $y_i = y(x_i)$, $i = 0, \dots, n-1$, assumed to come from a smooth function. Try a polynomial:

$$y(x_i) = \tilde{y}_i + \epsilon_i = \sum_{j=0}^{p-1} \theta_j x_i^j + \epsilon_i.$$

Writing out all n equations,

$$y_0 = \theta_0 + \theta_1 x_0 + \theta_2 x_0^2 + \dots + \theta_{p-1} x_0^{p-1} + \epsilon_0$$

$$y_1 = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \dots + \theta_{p-1} x_1^{p-1} + \epsilon_1$$

$\vdots$

$$y_{n-1} = \theta_0 + \theta_1 x_{n-1} + \dots + \theta_{p-1} x_{n-1}^{p-1} + \epsilon_{n-1},$$

that is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^{p-1} \\ 1 & x_1 & x_1^2 & \dots & x_1^{p-1} \\ \vdots & & & & \vdots \\ 1 & x_{n-1} & x_{n-1}^2 & \dots & x_{n-1}^{p-1} \end{bmatrix},$$

the *Vandermonde* matrix. Known: $\mathbf{y}$ and $\mathbf{X}$. Unknown: $\boldsymbol{\theta}$ (and $\boldsymbol{\epsilon}$).

“Linear” refers to the parameters, not to the features

Nothing forces the columns to be powers of x . Any set of basis functions works,

$$\tilde{y}_i = \sum_{j=0}^{p-1} \theta_j \phi_j(x_i), \quad \phi_j \in \{\text{polynomials, } \cos(jx), \sin(jx), \text{ splines, radial functions, } \dots\},$$

a *basis expansion*, and everything in this chapter applies verbatim.

- ▶ The number of columns p is the complexity knob. Degree d in one variable: $p = d + 1$. Degree d in two variables ($x^a y^b$, $a + b \leq d$): $p = (d + 1)(d + 2)/2$ – 21 parameters at $d = 5$, 66 at $d = 10$.
- ▶ The first column of ones has a name: its coefficient θ_0 is the *intercept*. It will need special treatment when we penalise.
- ▶ The Vandermonde matrix is horribly conditioned; the choice of basis is a numerical decision as well as a modelling one.

Pause and think 2: what is linear here?

Question

Which of the following are *linear regression* problems in the sense of this lecture?

1. $y = \theta_0 + \theta_1 x + \theta_2 x^2$
2. $y = \theta_0 e^{\theta_1 x}$
3. $y = \theta_0 \sin x + \theta_1 \cos 3x$
4. $y = \theta_0 / (1 + \theta_1 x)$
5. $\log y = \theta_0 + \theta_1 x$

Pause and think 2: what is linear here?

Question

Which of the following are *linear regression* problems in the sense of this lecture?

1. $y = \theta_0 + \theta_1 x + \theta_2 x^2$
2. $y = \theta_0 e^{\theta_1 x}$
3. $y = \theta_0 \sin x + \theta_1 \cos 3x$
4. $y = \theta_0 / (1 + \theta_1 x)$
5. $\log y = \theta_0 + \theta_1 x$

Answer

1, 3 and 5 are linear in θ (5 after taking the logarithm of the data – but note that this changes what “least squares” means, since the noise is now assumed additive on $\log y$). 2 and 4 are not: θ_1 appears inside a non-linear function. Number 2 becomes linear after a logarithm, number 4 does not.

Optimising the parameters: the cost function

Prediction $\tilde{\mathbf{y}} = \mathbf{X}\boldsymbol{\theta}$. Measure the spread between data and prediction by the *mean squared error*,

$$C(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \tilde{y}_i)^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2.$$

- ▶ This is one possible cost function – the one that follows from Gaussian noise, as we shall see shortly. Others exist (absolute error, χ^2 , ...).
- ▶ Some authors write $1/(2n)$ so that the factor 2 cancels on differentiation. It changes nothing about the minimiser – but keep track of such factors when comparing codes!

The optimisation problem is

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^p} \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2.$$

Minimising: the normal equations

Differentiate with respect to one parameter,

$$\frac{\partial C}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \left[\frac{1}{n} \sum_i \left(y_i - \theta_0 x_{i0} - \theta_1 x_{i1} - \cdots - \theta_{p-1} x_{i,p-1} \right)^2 \right] = -\frac{2}{n} \sum_i x_{ij} \left(y_i - \sum_k \theta_k x_{ik} \right) = 0,$$

and collect all p equations into one vector equation:

$$\frac{\partial C}{\partial \boldsymbol{\theta}} = -\frac{2}{n} \mathbf{X}^T (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) = 0 \quad \Rightarrow \quad \boxed{\mathbf{X}^T \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^T \mathbf{y}}$$

the *normal equations*. If $\mathbf{X}^T \mathbf{X}$ is invertible,

$$\boxed{\hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

The Hessian is $\frac{2}{n} \mathbf{X}^T \mathbf{X}$, positive semi-definite ($\mathbf{z}^T \mathbf{X}^T \mathbf{X} \mathbf{z} = \|\mathbf{X} \mathbf{z}\|^2 \geq 0$), so the cost is convex and this stationary point is the *global* minimum.

Pause and think 3: when does the inverse exist?

Question

$\mathbf{X}^T \mathbf{X}$ is $p \times p$. When is it invertible? Think about (a) two identical columns in $\mathbf{X}$, (b) more features than data points, $p > n$, (c) the degree-5 design matrix with columns x^4 and x^5 over $x \in [0, 1]$ – sketch the two functions.

Pause and think 3: when does the inverse exist?

Question

$\mathbf{X}^T \mathbf{X}$ is $p \times p$. When is it invertible? Think about (a) two identical columns in $\mathbf{X}$, (b) more features than data points, $p > n$, (c) the degree-5 design matrix with columns x^4 and x^5 over $x \in [0, 1]$ – sketch the two functions.

Answer

$\mathbf{X}^T \mathbf{X}$ is invertible if and only if the columns of $\mathbf{X}$ are linearly independent, i.e. $\mathbf{X}$ has full column rank p . (a) Two identical columns: singular, and any $\theta_j \leftrightarrow \theta_k$ trade-off gives the same fit – the parameters are not unique but the *prediction* $\mathbf{X}\hat{\boldsymbol{\theta}}$ is. (b) $p > n$: rank at most $n < p$, always singular; there are infinitely many exact interpolants. (c) Not singular, but x^4 and x^5 look almost alike on $[0, 1]$ (both near zero, then rising steeply), so the columns are strongly correlated and $\mathbf{X}^T \mathbf{X}$ is nearly singular: the inverse exists and is numerically terrible. The higher the degree, the worse – that is the reason for the SVD a few slides on.

The geometry: least squares is a projection

$\mathbf{X}\boldsymbol{\theta}$ ranges over the *column space* of $\mathbf{X}$ as $\boldsymbol{\theta}$ varies. Least squares asks for the point of that subspace closest to $\mathbf{y}$: the orthogonal projection.

$$\tilde{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (\text{the } \textit{hat matrix}),$$

symmetric, idempotent ($\mathbf{H}^2 = \mathbf{H}$), with $\text{Tr}(\mathbf{H}) = p$.

The normal equations read $\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}) = 0$:

At the optimum the residual $\boldsymbol{\epsilon} = \mathbf{y} - \tilde{\mathbf{y}}$ is *orthogonal to every column* of the design matrix. The fit has extracted from $\mathbf{y}$ everything that lies in the span of the features and left the rest untouched.

Pythagoras then gives $\|\mathbf{y}\|^2 = \|\tilde{\mathbf{y}}\|^2 + \|\boldsymbol{\epsilon}\|^2$ – the seed of R^2 and of the whole statistics of the fit.

Pause and think 4: the hat matrix

Question

Without computing anything: (a) what is $\mathbf{H}\mathbf{y}$ if $\mathbf{y}$ already lies in the column space of $\mathbf{X}$? (b) what is $\mathbf{H}\mathbf{X}$? (c) why must $\mathbf{H}^2 = \mathbf{H}$? (d) what does $\text{Tr}(\mathbf{H}) = p$ mean about the residuals – how many “directions” are left for them?

Pause and think 4: the hat matrix

Question

Without computing anything: (a) what is $\mathbf{H}\mathbf{y}$ if $\mathbf{y}$ already lies in the column space of $\mathbf{X}$? (b) what is $\mathbf{H}\mathbf{X}$? (c) why must $\mathbf{H}^2 = \mathbf{H}$? (d) what does $\text{Tr}(\mathbf{H}) = p$ mean about the residuals – how many “directions” are left for them?

Answer

(a) $\mathbf{H}\mathbf{y} = \mathbf{y}$: projecting a vector already in the subspace changes nothing. (b) $\mathbf{H}\mathbf{X} = \mathbf{X}$, each column is in the column space. (c) Projecting twice is projecting once. (d) $\mathbf{H}$ has p eigenvalues equal to 1 and $n - p$ equal to 0; the residual $(\mathbf{I} - \mathbf{H})\mathbf{y}$ lives in the $(n - p)$ -dimensional complement – the fit “uses up” p degrees of freedom, which is why $\hat{\sigma}^2$ will divide by $n - p$.

Optimality without calculus

Let $\hat{\theta}$ solve the normal equations. Then for every θ ,

$$\|\mathbf{y} - \mathbf{X}\theta\|_2^2 = \underbrace{\|\mathbf{y} - \mathbf{X}\hat{\theta}\|_2^2}_{\text{residual sum of squares}} + \underbrace{(\theta - \hat{\theta})^T \mathbf{X}^T \mathbf{X} (\theta - \hat{\theta})}_{\geq 0}.$$

Proof: write $\mathbf{y} - \mathbf{X}\theta = (\mathbf{y} - \mathbf{X}\hat{\theta}) - \mathbf{X}(\theta - \hat{\theta})$, expand the square; the cross term is $-2(\theta - \hat{\theta})^T \mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\theta}) = 0$ by the normal equations.

Three consequences, all reused later:

1. $\hat{\theta}$ is a *global* minimiser – no second derivatives needed.
2. It is unique iff $\mathbf{X}^T \mathbf{X}$ is positive definite (full column rank); otherwise the parameters are not unique but the fit $\mathbf{X}\hat{\theta}$ is.
3. The cost rises away from the optimum steeply in directions of large singular value of $\mathbf{X}$ and hardly at all in directions of small ones: a flat valley = a direction the data do not determine.

How to compute it: never form the inverse

$\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is a statement, not an algorithm. Forming $\mathbf{X}^T \mathbf{X}$ *squares the condition number*: $\kappa_2(\mathbf{X}^T \mathbf{X}) = \kappa_2(\mathbf{X})^2$. A degree-12 polynomial on $[0, 1]$: $\kappa(\mathbf{X}) \sim 10^8$, $\kappa(\mathbf{X}^T \mathbf{X}) \sim 10^{16}$ – every digit gone in double precision.

The stable routes are QR or the SVD, $\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$, through the pseudoinverse

$$\hat{\theta} = \mathbf{X}^+ \mathbf{y} = \sum_i \frac{\mathbf{u}_i^T \mathbf{y}}{\sigma_i} \mathbf{v}_i,$$

which also handles the rank-deficient case (minimum-norm solution when $p > n$).

```
def ols(X, y):
    """OLS through the SVD -- never the normal equations."""
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    return Vt.T @ (U.T @ y / s)

def ols_normal_equations(X, y):
    """The textbook formula. Shown for comparison; do not use it."""
    return np.linalg.pinv(X.T @ X) @ X.T @ y
# or simply: np.linalg.lstsq(X, y, rcond=None)[0]
```

Almost everything in the rest of the chapter is a statement about the factors $1/\sigma_i$.

Own code for the degree-5 fit

```
# matrix inversion to find theta (fine for a well-conditioned 6x6 problem)
theta = np.linalg.inv(X.T @ X) @ X.T @ y
ytilde = X @ theta
# or the least-squares functionality in numpy (SVD based -- preferred)
theta_np = np.linalg.lstsq(X, y, rcond=None)[0]

def R2(y_data, y_model):
    return 1 - np.sum((y_data - y_model)**2) / np.sum((y_data - np.mean(y_data))**2)

def MSE(y_data, y_model):
    return np.sum((y_data - y_model)**2) / np.size(y_model)

def RelativeError(y_data, y_model):
    return abs((y_data - y_model) / y_data)

print(theta) # six coefficients
print(R2(y, ytilde), MSE(y, ytilde))
```

- ▶ MSE: what we minimised, in squared units of the target – its absolute value says little on its own.
- ▶ $R^2 = 1 - \sum (y_i - \tilde{y}_i)^2 / \sum (y_i - \bar{y})^2$: one minus the fraction of variance left unexplained; 1 is perfect, 0 is “no better than the mean”, negative is possible on test data.
- ▶ Relative error: natural when targets span decades, dangerous near $y_i = 0$.

Pause and think 5: what R^2 does and does not measure

Question

On the *training* data, what happens to R^2 when we add one more column to $\mathbf{X}$ – any column, even pure noise? What is R^2 when $p = n$? So what does a training R^2 of 0.999 tell us?

Pause and think 5: what R^2 does and does not measure

Question

On the *training* data, what happens to R^2 when we add one more column to $\mathbf{X}$ – any column, even pure noise? What is R^2 when $p = n$? So what does a training R^2 of 0.999 tell us?

Answer

It can only rise: every θ available before is still available (with a zero in the new place), so the minimum RSS cannot increase. At $p = n$ the column space is all of $\mathbb{R}^n$, $\mathbf{H} = \mathbf{I}$, $\text{RSS} = 0$ and $R^2 = 1$ regardless of what the columns contain. A training R^2 near one therefore says nothing about predictive power; only data the model has not seen do. Every number we report in this course is a *test* quantity unless we say otherwise.

Weighted least squares: the χ^2 function

Measurements come with error bars. A point known to 10% should not constrain the fit as strongly as a point known to 1%. With the standard deviation σ_i of measurement i ,

$$\chi^2(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=0}^{n-1} \frac{(y_i - \tilde{y}_i)^2}{\sigma_i^2} = \frac{1}{n} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-2} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}), \quad \boldsymbol{\Sigma} = \text{diag}(\sigma_0, \dots, \sigma_{n-1}).$$

Every residual is measured in units of its own uncertainty. Setting the gradient to zero,

$$\frac{\partial \chi^2}{\partial \boldsymbol{\theta}} = -\frac{2}{n} \mathbf{X}^T \boldsymbol{\Sigma}^{-2} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) = 0 \implies \hat{\boldsymbol{\theta}} = (\mathbf{X}^T \boldsymbol{\Sigma}^{-2} \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\Sigma}^{-2} \mathbf{y}.$$

Define $\mathbf{A} = \boldsymbol{\Sigma}^{-1} \mathbf{X}$ ($a_{ij} = x_{ij}/\sigma_i$) and $\mathbf{b} = \boldsymbol{\Sigma}^{-1} \mathbf{y}$ ($b_i = y_i/\sigma_i$): then $\hat{\boldsymbol{\theta}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$ – *weighted least squares is ordinary least squares on rescaled data.*

Pause and think 6: what OLS silently assumes

Question

Setting all σ_i equal recovers OLS. What assumption about the noise does OLS therefore make? If one data point has a *very* large σ_i , what does the weighted fit do with it? And if the error bars are all wrong by the same factor c , does the fit change?

Pause and think 6: what OLS silently assumes

Question

Setting all σ_i equal recovers OLS. What assumption about the noise does OLS therefore make? If one data point has a *very* large σ_i , what does the weighted fit do with it? And if the error bars are all wrong by the same factor c , does the fit change?

Answer

OLS assumes *homoscedastic*, i.e. equal-variance, noise. A point with huge σ_i gets weight $1/\sigma_i^2 \rightarrow 0$: it is effectively dropped. A common factor c scales χ^2 by $1/c^2$ but leaves the minimiser unchanged – the *relative* weights are what matter for $\hat{\theta}$; the absolute scale matters for the uncertainties and for the value of $\chi_{\min}^2$.

The χ^2 function for a straight line: no matrix to invert

For $y \approx \theta_0 + \theta_1 x$ the two equations $\partial\chi^2/\partial\theta_{0,1} = 0$ read

$$\sum_i \frac{y_i - \theta_0 - \theta_1 x_i}{\sigma_i^2} = 0, \quad \sum_i x_i \frac{y_i - \theta_0 - \theta_1 x_i}{\sigma_i^2} = 0.$$

With the sums

$$\gamma = \sum_i \sigma_i^{-2}, \quad \gamma_x = \sum_i x_i \sigma_i^{-2}, \quad \gamma_y = \sum_i y_i \sigma_i^{-2}, \quad \gamma_{xx} = \sum_i x_i^2 \sigma_i^{-2}, \quad \gamma_{xy} = \sum_i x_i y_i \sigma_i^{-2},$$

$$\theta_0 = \frac{\gamma_{xx}\gamma_y - \gamma_x\gamma_{xy}}{\gamma\gamma_{xx} - \gamma_x^2}, \quad \theta_1 = \frac{\gamma\gamma_{xy} - \gamma_x\gamma_y}{\gamma\gamma_{xx} - \gamma_x^2}.$$

- ▶ This is Cramer's rule on the 2×2 normal equations; the denominator is $\det(\mathbf{A}^T \mathbf{A})$.
- ▶ Uncertainties: with $\mathbf{H} = (\mathbf{A}^T \mathbf{A})^{-1}$ (matrix elements h_{jk}), propagating $\sigma^2(\theta_j) = \sum_i \sigma_i^2 (\partial\theta_j/\partial y_i)^2$ gives simply $\sigma^2(\theta_j) = h_{jj}$ – the diagonal of the inverse. (Exercise this week: derive it.)

Pause and think 7: reading χ^2 at the minimum

Question

A physicist reports $\chi_{\min}^2$ together with the fitted parameters. If the model and the σ_i are both right, roughly what value do you expect for $\sum_i (y_i - \tilde{y}_i)^2 / \sigma_i^2$ at the minimum, for n points and p parameters? What would a value ten times larger suggest? Ten times smaller?

Pause and think 7: reading χ^2 at the minimum

Question

A physicist reports $\chi^2_{\min}$ together with the fitted parameters. If the model and the σ_i are both right, roughly what value do you expect for $\sum_i (y_i - \tilde{y}_i)^2 / \sigma_i^2$ at the minimum, for n points and p parameters? What would a value ten times larger suggest? Ten times smaller?

Answer

About $n - p$: each standardised residual has variance ≈ 1 and the fit removes p degrees of freedom (Section 3.4 of the book proves $\mathbb{E}[\sum (y_i - \tilde{y}_i)^2 / \sigma_i^2] = n - p$). Ten times larger: the model does not describe the data at the stated precision (or the error bars are underestimated). Ten times smaller: the error bars are overestimated – the fit is “too good”. The *reduced* χ^2 , $\chi^2_{\min} / (n - p)$, should be close to one.

Where does the squared error come from? Maximum likelihood

Assume the data are generated by $y_i = \mathbf{X}_{i,*}\boldsymbol{\theta} + \varepsilon_i$ with independent Gaussian noise, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Then

$$p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \mathbf{X}_{i,*}\boldsymbol{\theta})^2}{2\sigma^2}\right], \quad p(\mathbf{D} \mid \boldsymbol{\theta}) = \prod_{i=0}^{n-1} p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}).$$

Maximum likelihood: choose $\boldsymbol{\theta}$ to make the observed data most probable. Take $-\log$ (products $\rightarrow$ sums; no underflow):

$$C(\boldsymbol{\theta}) = -\log p(\mathbf{D} \mid \boldsymbol{\theta}) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2}{2\sigma^2}.$$

The first term does not depend on $\boldsymbol{\theta}$; the second is the least-squares cost. **Least squares is maximum likelihood for Gaussian noise.**

The useful direction of the statement

A different noise model gives a different loss. Specify the distribution of the target and the loss function is *determined* – for regression, for logistic regression, for neural networks.

Pause and think 8: a different noise model

Question

Suppose the noise is Laplace-distributed, $p(\varepsilon) \propto e^{-|\varepsilon|/b}$, with heavier tails than the Gaussian. Repeat the argument in your head: which cost function results? Which of the two fits is more robust to a handful of outliers, and why?

Pause and think 8: a different noise model

Question

Suppose the noise is Laplace-distributed, $p(\varepsilon) \propto e^{-|\varepsilon|/b}$, with heavier tails than the Gaussian. Repeat the argument in your head: which cost function results? Which of the two fits is more robust to a handful of outliers, and why?

Answer

$-\log p$ gives $\sum_i |y_i - \tilde{y}_i|/b$: the *mean absolute error*. Squaring gives a point ten standard deviations out a hundred times the weight of one at a single standard deviation; the absolute value gives it ten times. So the MAE fit is far less dominated by outliers. Choosing squared error for heavy-tailed data is not a matter of taste but a modelling error.

Statistical properties of the least-squares estimator

$\hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is a function of the random $\mathbf{y}$, hence itself random. With $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$, $\mathbb{E}[\boldsymbol{\varepsilon}] = 0$, $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$:

$$\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta} \quad (\text{unbiased, if the linear model is right}), \quad \text{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}.$$

The covariance is the inverse Hessian times the noise variance. σ^2 is estimated from the residuals,

$$\hat{\sigma}^2 = \frac{\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}\|_2^2}{n - p}, \quad \mathbb{E}[\hat{\sigma}^2] = \sigma^2 \quad (\text{because } \mathbf{e} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}, \text{Tr}(\mathbf{I} - \mathbf{H}) = n - p),$$

and a confidence interval for coefficient j is

$$\hat{\theta}_j \pm t_{\alpha/2, n-p} \hat{\sigma} \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}} \quad (t \rightarrow 1.96 \text{ for } 95\%, \ n - p \text{ large}).$$

Under Gaussian noise these are exact: $\hat{\boldsymbol{\theta}}$ is Gaussian, independent of the residuals, and $(n - p)\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$ – which is also the reduced- χ^2 rule from earlier.

Reading the variance through the SVD: what collinearity does

With $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, $\mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^T$ and

$$\hat{\boldsymbol{\theta}}_{\text{OLS}} = \sum_i \frac{\mathbf{u}_i^T \mathbf{y}}{\sigma_i} \mathbf{v}_i, \quad \text{Var}(\hat{\boldsymbol{\theta}}_{\text{OLS}}) = \sigma^2 \sum_i \frac{\mathbf{v}_i \mathbf{v}_i^T}{\sigma_i^2}.$$

Along the i th right singular direction the coefficient is obtained by dividing by σ_i and its variance by dividing by σ_i^2 .

A direction in feature space along which the data barely vary has a small singular value. The coefficient along it is large and its variance enormous: the fit becomes a huge positive coefficient on one feature cancelling a huge negative one on another, and the pair swings wildly when the data are perturbed. Nothing is wrong with the algebra – the data simply do not determine that combination.

This is the same κ that ruined the numerics earlier, now ruining the statistics.

Pause and think 9: two almost identical features

Question

Two columns of $\mathbf{X}$ are nearly proportional, say temperature in $^{\circ}\text{C}$ and in K (plus rounding). Predict, before computing: what happens to the two fitted coefficients? To their standard errors? To the *fitted values* $\tilde{\mathbf{y}}$? And to predictions at a new point that respects the same near-proportionality?

Pause and think 9: two almost identical features

Question

Two columns of $\mathbf{X}$ are nearly proportional, say temperature in °C and in K (plus rounding). Predict, before computing: what happens to the two fitted coefficients? To their standard errors? To the *fitted values* $\tilde{\mathbf{y}}$? And to predictions at a new point that respects the same near-proportionality?

Answer

The individual coefficients are wildly uncertain – large, opposite in sign, and unstable under resampling; their standard errors blow up as $\sigma/\sigma_{\min}$. The fitted values and predictions at points obeying the same relation are perfectly stable: they depend only on the sum of the two contributions, which the data determine well. Only when a new point breaks the relation (a temperature that is °C on one column and something else on the other) does the instability reach the prediction. Collinearity is a problem of *interpretation and extrapolation*; Ridge regression, next time, is its cure.

Summary, and what comes next

- ▶ A linear model is linear in θ ; the columns of $\mathbf{X}$ can be anything – powers, Fourier modes, splines. p is the complexity knob.
- ▶ Minimising the MSE gives the normal equations $\mathbf{X}^T \mathbf{X} \theta = \mathbf{X}^T \mathbf{y}$ and, for full column rank, $\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$: convex, global, unique.
- ▶ Geometrically it is the projection $\mathbf{H}\mathbf{y}$ onto the column space; the residual is orthogonal to every feature; $\text{Tr } \mathbf{H} = p$.
- ▶ Numerically: solve by SVD or QR, never by inverting $\mathbf{X}^T \mathbf{X}$ (condition number squared).
- ▶ Error bars go in as weights: χ^2 is OLS on $\mathbf{X}/\sigma$, $\mathbf{y}/\sigma$; the reduced χ^2 should be near one.
- ▶ MSE and R^2 measure quality – on *test* data.
- ▶ Least squares is maximum likelihood for Gaussian noise; another noise model gives another loss.
- ▶ $\hat{\theta}$ is unbiased with covariance $\sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$; through the SVD, the variance along a singular direction is σ^2/σ_i^2 – collinearity made quantitative.

Coming lectures

Gauss–Markov and the bias–variance trade-off; Ridge and Lasso regression as the cure for collinearity (Sections 3.7–3.10); then resampling, cross-validation and gradient descent.